വിജയഭേരി

മലപ്പുറം ജില്ലാ പഞ്ചായത്ത് വിദ്യാഭ്യാസപദ്ധതി



LEAP-26

ENGLISH MEDIUM

SSLC

BIOLOGY

QUESTION POOL

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EDUCATIONAL PROJECT

LEAP 26

SSLC BIOLOGY QUESTION POOL

അഭിമുഖം

2001–2002 വർഷത്തിൽ മലപ്പുറം ജില്ലയുടെ സമഗ്ര വിദ്യാഭ്യാസ പുരോഗതി ലക്ഷ്യം വെച്ചു മലപ്പുറം ജില്ലാ പഞ്ചായത്ത് ആസൂത്രണം ചെയ്തു നടപ്പിലാക്കിവരുന്ന പദ്ധതിയാണ് വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതി.
കഴിഞ്ഞ 24 വർഷത്തിനുള്ളിൽ ജില്ലയുടെ വിദ്യാഭ്യാസ മുഖച്ഛായ മാറ്റാൻ ഈ പദ്ധതിയിലൂടെ കഴിഞ്ഞിട്ടുണ്ട്. എസ്എസ്എൽസി വിജയശതമാനത്തിലും പ്ലസ് ടു വിജയശതമാനത്തിലും മത്സര പരീക്ഷകളിലും ഏറെ പിന്നിലായിരുന്ന ജില്ല ഇന്ന് സംസ്ഥാനത്തെ മറ്റേത് ജില്ലയോടും കിടപിടിക്കാവുന്ന രീതിയിൽ മാത്രമല്ല, പല മേഖലയിലും അവരെക്കാൾ മുന്നേറാൻ ഇന്ന് ജില്ലയ്ക്ക് കഴിഞ്ഞിട്ടുണ്ട്.
ഏറ്റവും കൂടുതൽ എ പ്ലസ്സുകളുള്ള ജില്ല, ഏറ്റവും കൂടുതൽ കുട്ടികൾ കേന്ദ്ര സർവകലാശാലയിലേക്ക് പ്രവേശനം നേടുന്ന ജില്ല, മെഡിക്കൽ എൻജിനീയറിങ് എൻട്രൻസ് പരീക്ഷകളിൽ മുന്നിൽ നിൽക്കുന്ന ജില്ല ഇന്ന് മലപ്പുറം ജില്ലയാണ്. വിദ്യാഭ്യാസപരമായി പിന്നാക്കം നിൽക്കുന്ന ജില്ല എന്ന അപഖ്യാതി നമ്മളിന് മറികടന്നു.

മറ്റ ജില്ലകളിൽ നിന്ന് ഭിന്നമായി ജില്ലയിലെ എല്ലാ തദ്ദേശ സ്വയംഭരണ സ്ഥാപനങ്ങൾക്കും എംഎൽഎമാർക്കും ഇന്ന് പ്രത്യേക വിദ്യാഭ്യാസ പദ്ധതികളുണ്ട്. ഇത്തരം എല്ലാ പദ്ധതികൾക്കും പ്രചോദനമായത് വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതിയാണ്. സംസ്ഥാനതലത്തിൽ മാത്രമല്ല ദേശീയ തലത്തിൽ പോലും ഈ പദ്ധതി ചർച്ച ചെയ്യപ്പെട്ട വരുന്നു. യൂണിവേഴ്സിറ്റികളിൽ വിജയിഭേരിപദ്ധതി ഗവേഷണ വിദ്യാർത്ഥികൾക്ക് പഠന വിഷയമാണ്.

ഈ പദ്ധതിയുടെ വിജയം മലപ്പുറം ജില്ലാ പഞ്ചായത്തിന്റെ വിജയമല്ല, ഇത് ഏറ്റെടുത്ത അധ്യാപകരുടെയും രക്ഷിതാക്കളുടെയും വിദ്യാർത്ഥികളുടെയും പിന്തുണയാണ് ഈ പദ്ധതിയുടെ വിജയത്തിന്റെ അടിസ്ഥാനം. അതുകൊണ്ടുതന്നെ അവരോടുള്ള കടപ്പാട് ഈ അവസരത്തിൽ പ്രത്യേകം രേഖപ്പെടുത്തുന്നു.

കഴിഞ്ഞ വർഷങ്ങളിലേതുപോലെ ഈ വർഷം നിലവിൽ വന്ന പുതിയ ജില്ലാ പഞ്ചായത്ത് ബോർഡും വിജയഭേരി പദ്ധതി കൂടുതൽ ഊർജ്ജസ്വലമായി മുന്നോട്ടുകൊണ്ടുപോകാൻ തീരുമാനിച്ചിട്ടുണ്ട്. അതിന്റെ ആദ്യപടിയായി ഈ വർഷം എസ്എസ്എൽസി പരീക്ഷ എഴുതുന്ന വിദ്യാർത്ഥികൾക്കും അധ്യാപകർക്കും സഹായകരമായ രീതിയിൽ എല്ലാ വിഷയങ്ങളുടെയും ചോദ്യശേഖരം പുറത്തിറക്കുകയാണ്.

പുതിയ പാഠപുസ്തകങ്ങളും പുതിയ രീതിയിലുള്ള പരീക്ഷ രീതിയുമാണ് ഈ വർഷത്തെ വിദ്യാർഥികൾ അഭിമുഖീകരിക്കേണ്ടത്. അവർക്ക് ഏറെ സഹായകമായ രീതിയിൽ റിവിഷൻ നടത്തുന്നതിനും പുതിയ രീതിയിലുള്ള ചോദ്യങ്ങൾ പരിചയപ്പെടുത്തുന്നതിനുമാണ് ഇത്തരമൊരു ചോദ്യശേഖരം തയ്യാറാക്കിയിരിക്കുന്നത്. എല്ലാ വിഷയങ്ങളുടെയും എല്ലാ പാഠങ്ങളിലുമുള്ള പ്രതീക്ഷിത ചോദ്യങ്ങൾ ഇതിൽ ലഭ്യമാക്കിയിട്ടുണ്ട്. ഈ ചോദ്യ ശേഖരം വിദ്യാർത്ഥികൾക്കും അധ്യാപകർക്കും ഏറെ സഹായകരമായിരിക്കുമെന്ന് പ്രതീക്ഷിക്കുന്നു.

ദൈനംദിന പ്രവർത്തനങ്ങൾക്കിടയിൽ ഇത്തരമൊരു ഭാരിച്ച ജോലി ഏറ്റെടുത്ത് കൃത്യസമയത്ത് പൂർത്തീകരിച്ച് പ്രസിദ്ധീകരിക്കുന്നതിന് രാപ്പകലോടൊപ്പം നിന്ന അധ്യാപകരെ പ്രത്യേകം അഭിനന്ദിക്കുന്നു.

പി എ ജബ്ബാർ ഹാജി
പ്രസിഡന്റ്
മലപ്പുറം ജില്ലാ പഞ്ചായത്ത്

അഡ്വ. എ.പി. സുജിനി
വൈസ് പ്രസിഡൻറ്

പി കെ അസ്സു
ആരോഗ്യ-വിദ്യാഭ്യാസ സ്റ്റാൻഡിംഗ് കമ്മിറ്റി ചെയർമാൻ

ടി സലിം
കോഡിനേറ്റർ
വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതി

BIOLOGY CHAPTERS

1. GENETICS OF LIFE
2. PATHS OF EVOLUTION
3. BEHIND SENSATIONS
4. CHEMORECEPTION IN ORGANISMS
5. IMMUNITY AND HEALTHCARE
6. BIOLOGY AND TECHNOLOGY

SSLC BIOLOGY QUESTION POOL

1. GENETICS OF LIFE

1. Fill up the blanks by observing the relationship between the first pair.
   a. DNA: Two strands:: RNA: …….
   b. Associated with ribosome: rRNA :: Carry amino acids to the ibosome:……
   c. TT: tall:: tt: …….
   d.Incomplete Dominance: A dominant allele cannot fully hide the allele of the recessive trait :: Co-dominance :……?

2. The genetic constitution of males.
   a. 44 + XX b. 44 + XY c. 43 + XX d. 43 + XY

3. The sugar seen in RNA.
   a. Ribose b. Ribosome c. Ribulose d. Deoxyribose

4.The genetic constitution of sperm.
   a. 44 + XX b. 44 + XY c. 22 + X d. 22 + Y

5. Which of the following is the correct nitrogen base pair in DNA?
   a. Adenine-Guanine b. Adenine-Thymin c. Cytosine-Thymine d. Thymine-Guanin

6.Hybridisation of two contrasting traits of the characteris illustrated below. Which will be the recessive trait of the F1 generation.

   Tall (TT), Dwarf (tt), Tall (Tt), Dwarf (Tt)

7. Which among the following is not a component of a nucleotide?
   a. Phosphate c. Protein b.Sugar d.Nitrogen base

8. Which of the following nucleotides is found only in RNA?
   A U C T

9. Select the correct answer.
   Statement 1: There are 22 somatic chromosomes in humans.
   Statement 2: The sex chromosomes in male is XY.
   a. Statement 1 and statement 2 are correct.
   b. Statement 1 and statement 2 are wrong.
   c. Statement 1 wrong and statement 2 correct.
   d. Statement 1 correct and statement 2 wrong

10. a. What is formed when DNA strands wrap around a histone octamer? How does this become a chromosome?
    b. Draw a diagram of a chromosome and label the centromere and chromatids.

11. Complete the flow chart showing gene action.
    mRNA is formed from DNA
    ↓
    (a)
    ↓
    mRNA reaches outside the nucleus
    ↓
    (b)
    ↓
    (c)
    ↓
    Protein is synthesized

12. The note prepared bySURYA on Mendel's inferences during the classroom analysis of Mendel's hybridization experiment in pea plants. based on a single trait is given below. Analyse the statements in the note and correct those that are wrong ones.
    a. A trait is controlled by a specific factor.
    b. A character is expressed and the other remains hidden in the first generation.
    c. The character that remains hidden in the first generation does not appear in the second generation.
    d. The ratio of characters in the second generation is 3:1.

13. Complete the flowchart illustrating the location of gene by using the information given in the box:
    (nucleus, gene, DNA, cell, chromosome)

14. Gene itself is an allele. allele itself is a gene. Substantiae your answer.

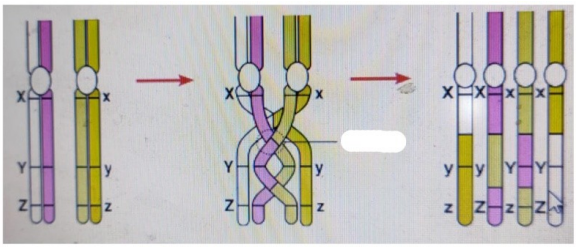
15. Analyse the given genetic makeup and answer the questions.
    44+XY
        a. Give the number of somatic chromosomes.
        b. Is the individual with this genetic makeup male or female? Why?

16. Analyse the statement and answer the questions.
    '*A sudden heritable change in the genetic constitution of an organism also results in variations*'.
    a. Identify and name the process.
    b. Write the conditions which leads to this.

17. Observe the the statement against color discrimination and answer the questions.
    - Stop Racism
    - Colour difference is due to the rise or fall in the production of pigment in the skin.
    - Skin colour is not racial. Biologically, we all are of the same race.

    a. Name the pigment mentioned for the skin colour.
    b. What is the reason for the rise and fall in the production of this pigment?

18. Analyse the illustration and answer the questions.



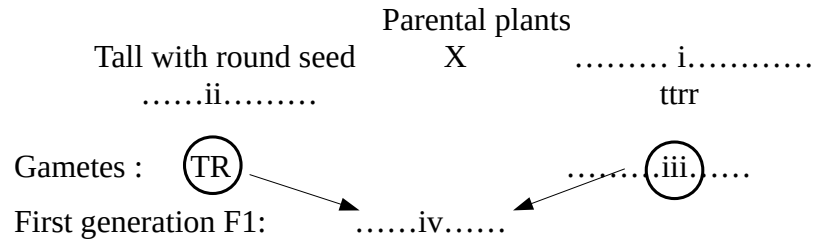
    a. Identify the process.
    h. What is the importance of this process?

19.Draw a DNA nucleotide and label its molecules.

20. Analyse each of the given situations and explain how variations are formed in the offspring:
    a. During meiosis. b. Gamete formation.

21. Select the statements related to DNA & RNA from the statements given below.
    a. Two strands.
    b. Deoxyribose sugar is seen.
    c. The nitrogen bases namely Adenine, Guanine, Uracil and Cytosine is seen.
    d. Deoxyribonucleic acid,
    e. Single stranded.
    f. The nitrogen bases namely Adenine, Guanine, Thymine and Cytosine is seen.
    g. Ribo nucleic acid.
    h. Ribose sugar is seen.

22. Complete the i, ii, iii and iv in the illustration.



23.Write reason:

a. ABO blood group.
b. Roan coat in cattle's.

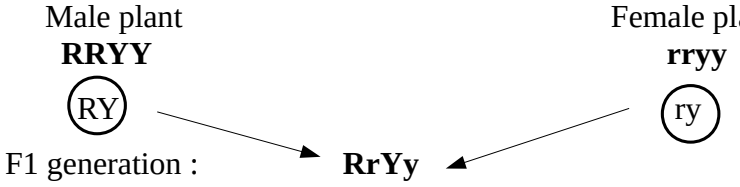
24. The stages related to action of gene is given below. Arrange them in correct order.

a. mRNA reaches the ribosome.
b. mRNA reaches outside the nucleus.
c. Protein is synthesized.
d. IRNA carries amino acids to the ribosome.
e. mRNA forms from DNA.
f. Based on the information in mRNA amino acids are added.

25. Select the correct statements.

a. 46 chromosomes are seen in humans.
b. 23 autosomes are found in humans
c. Genetic constitution of male is 44 +XX
d. Sex chromosomes are of two types.
e. Genetic constitution of female is 46+XY

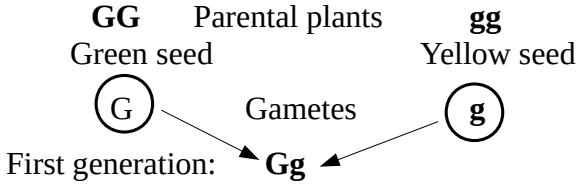
26. Observe the illustration of hybridization of two types of pea plants and answer the questions.



R-Round seed **r**-wringled seed **Y**-yellow seed **y**-green seeded .

a. Which are the characters expressed in first generation?
b. Which are the hidden characters in this experiment?
c. Write the possible expression of different characters in next generation.

27.Observe the illustration given below and answer the questions.



a Identify the dominant character.
b.How does the parental plant with green coloured seed and the plant in the first generation differ in their alleles?
c. Describe alleles.
d.why does the recessive trait in F1 appear in second generation?

28. Opinions of three students on protein synthesis is given below. Analyses it and answer the question.

Student 1: 'Protein synthesis begins in the cytoplasm and ends in the nucleus'.
Student 2: 'Protein synthesis begins in the nucleus and ends in the cytoplasm'.
Student 3: 'The entire process of protein synthesis takes place in the nucleus'.

Whose opinion is correct? Explain why?

29. Observe the illustration related to action of genes and answer the questions.



a. What is the function that takes place in the organelle marked as X in this process?
b. What is the importance of the other two RNAs that participate in this process
c. Explain the two stages of protein synthesis.

2. PATHS OF EVOLUTION

1. Choose the right answer from the barackets:
[Cerebrum, Cerebellum, Thalamus, Pons, Medulla oblongata]

a. Part that controls involuntary actions.
b. Part that evokes sensations.
c. Part that receives impulses from the vestibular nerve.

2. Which one is odd? Write the common characteristic of the others.
Gibbon, Monkey, Gorilla, Chimpanzee

3. **Statement** 1. The structure of Australopithecus confirms bipedalism.
Statement 2. Homo neanderthalensis had a small, sloping forehead and a thick eye brows.

a. Statement 1 and Statement 2 are correct,
b. Statement 1 and Statement 2 are incorrect.
c. Statement 1 is incorrect, Statement 2 is correct.
d. Statement 1 is correct, Statement 2 is incorrect.

4. Based on the following, explain how it substantiates evolution.

a. Palaeontology b. Physiology and Biochemistry

5. Explain Darwin's theory of evolution based on the given indicators.

* Over production
* Struggle for existence

6. **Man, Cat, Whale, Bat**.

a. What similarity do these organs have in their internal structure?
b. How does the comparative study of these organs provide evidence for evolution?

7. Arrange the following items appropriately under the headings: **Darwinism** and **Lamarckism**:

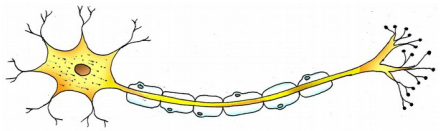
- Inheritance of variations
- Natural selection

- Only long necked giraffe survive
- Variation acquired in the life span
- Through continuoe use, giraffe’s neck elongates
- Use and disuse

8. The links in human evolution are given below. Arrange them according to brain size.

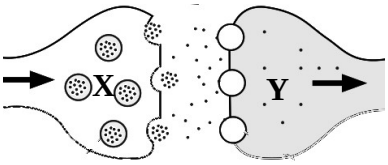
   Homo erectus, Homo sapiens, Homo neanderthalensis,
   Australopithecus, Sahelanthropus chadensis, Homo habilis.

9. Redraw the given figure and label the name of following parts on it:



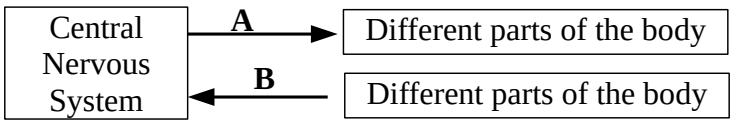
   a. Part that release neurotransmitter.
   b. The part that receives impulse from adjacent cell.
   c. Part that acts as an insulator.

10. Observe the illustration of impulse transmission through synapse and answer the following questions.



   a. Which part is denoted as 'X'?
   b. Which chemical is secreted from 'X'?
   c. Do impulse transmit from 'Y' to 'X? Why?

11. Analyse the illustration and answer the questions that follows:

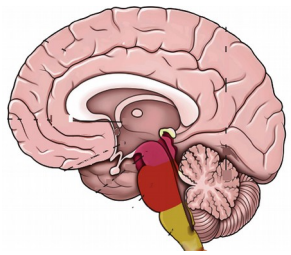


   a. Identify the nerves A and B.
   b. Are messages being exchanged between A and B? Explain.

12. Choose only the correct statements from the following:

   a. The central nervous system consists of the brain and the spinal cord
   b. The peripheral nervous system consists of 31 pairs of cranial nerves and 12 pairs of spinal nerves.
   c. The sympathetic and parasympathetic systems are part of the central nervous system.
   d. The autonomic nervous system, which is part of the peripheral nervous system, helps in overcoming emergency situations.
   e. The central nervous system consists of the brain and the spinal cord.

13. Redraw the diagram. Identify and label the parts according to the hints.



   a. Relay station of impulses.
   b. The part that controls heart beat.

14. The main parts and functions of brain are given below. Arrange them in the following table suitably.

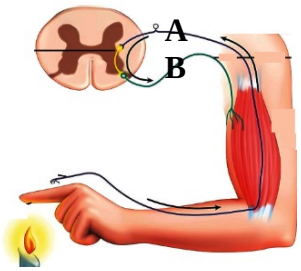
Thalamus, Controls voluntary movements, Cerebellum, Acts as relay station of impulses, Hypothalamus, Maintains equilibrium of the body, Cerebrum, Maintains homeostasis.

Part	Functions
Thalamus	

15. A table that includes the parts of spinal cord and their functions is given below. Arrange column B appropriately in accordance with the column B.

A. Part of spinal cord	**B**. Function/Peculiarity
Central canal	Transmits impulse to the spinal cord
White matter	Cell body of neurons seen here
Dorsal root	Fluid present here nourishes the spinal cord
Grey matter	Neurons with myelin sheath seen here

16. Analyze the illustration and answer the questions.



a. What is indicated in the illustration?
b. A and B are neurons. Write their difference.
c. Write an example related to this illustration.

17. Arrange the following statements suitably in the table given below. Give title to the table also.

a. Decreases production of saliva
b. Increases heartbeat
c. No direct influence on hormone production
d. Pupil of the eye constricts
e. Slows down peristalsis
f. Urinary bladder empties

Title:	Title:
* * *	* * *

3. BEHIND SENSATIONS

1. Analyse the table and findout the correct one from the given options.

A	B
P. Short-sightedness	i. Shortened eyeball
Q. Long sightedness	ii. Eye lens becomes opaque
R. Astigmatism	iii. Enlarged eyeball
	iv. Irregular cornea/lens curvature

a. p- iii, Q- i, R- iv b. p- i, Q- iii, R- ii

c. p- iv, Q- ii, R- i d. p- ii, Q- iv, R- iii

2. Identify the statements which are correct while viewing distant objects.

i. Ciliary muscles relaxes
ii. Lens curvature increases
iii. Ligaments stretches
iv. Ciliary muscles contracts

A. i and ii B. i and ii C. ii and iii B. ii and iv

3. Eye : Photoreceptors,

Tongue: ---------- ?

4. Name the organ where the following fluid seen. (Model: Brain: Cerebrospinal fluid)

- Endolymph
- Aqueous humor

5. Some changes taking place in the eye while exposed to dim light is given below. Analyse them and find out the correct statements.

i. Radial muscles contract
ii. Pupil dialates
iii. Circular muscles contract
iv. Pupil constricts

a. i and ii b. i and iv c. ii and iii d. iii and iv

6. Analyse the table and find out the correct one from the given options.

A	B
P. Amoeba	a. Jacobson’s organ
Q. Dog	b. Chemoreceptor
R. Snake	c. Olfactory receptors
	d. Eyespot

a. p- i, Q- ii, R- iv

b. p- ii, Q- iii, R- i

c. p- iv, Q- i, R- ii

d. p- iii, Q- iv, R- i

7.



Identify the photoreceptor. Name the pigment present in it.

8. A person is affected with damage to inner ear and lost balancing, But can hear well.

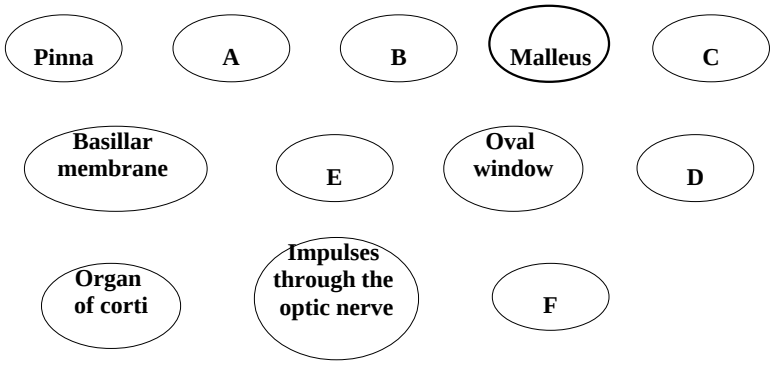
Give an explanation for this.

9. “Much attention should be given to noise pollution”. Do you agree with? Substantiate.

10. Complete the given flow chart

Olfactory molecules → ---------- → ---------- → ----------→Sense of smell.

11. Complete this and identify the process mentioned in the flow chart:



12. Redraw the diagram and label the following parts on it.



a. The part which is adjusted with the intensity of light

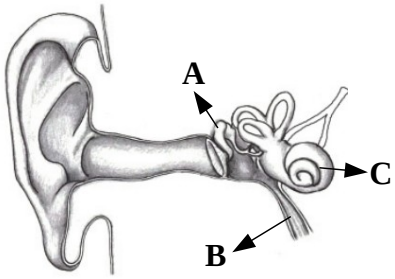
b. The chamber which is filled with jelly like substance

c. The muscles which help to alter the curvature of thelens

13. A person can not distinguish yellow colour clearly.

Name the disorder. What will be its reason? Name the cone cells related with this.

14. Observe the figure and answer the following questions:



a. Name the B and C of this figure.

b. Mention the function of B

c. Name the bones indicated as A.

d. Define Organ of Corti.

4. CHEMORECEPTION IN ORGANISMS

1. Find out the correct statement.

a) Gibberilins influences the growth of fruits and seeds.

b) Gibberilins stimulate cell division in both stem and roots.

c) Gibberilins break seed dormancy.

d) All of the above.

2. **Statement**: Photoperiodism take place in plants.

Reason: It is due the presence of phytochrome.

a) Statement correct, reason incorrect

b) Statement and reason correct

c) Statement and reason incorrect

d) Statement incorrect, reason correct

3. i : Gigantism occurs due to the overproduction of somatotropin after growth phase.

ii: Gigantism occurs due to the overproduction of somatotropin during growth phase.

a) (i) & (ii) correct b) (i) correct, (ii) incorrect

c) (i) & (ii) incorrect d) (i) incorrect, (ii) correct

4. Find out the correct statement.

a) Oxytocin is secreted by the posterior lobe of pituitary gland.

b) Oxytocin is secreted by the hypothalamus.

c) Oxytocin is secreted by the anterior lobe of pituitary gland.

d) None of the above.

5. Complete the following.

Releasing hormone → Pituitary → GTH → Target cell → Hormone-receptor complex

a). From where the releasing hormone produced?

b). Which is the target cell of GTH?

b). Why does a hormone only act at its own target cells?

7. One of the students opined that combined action of hormones and nervous system helps the body to overcome emergency situations.

a) Which hormone and which part of the nervous system are referred here?

b) Write any two combined actions of these.

8. Complete the illustration using the hints.

Hints: a & c are glands, b & d are hormones

----a------ produces -------b------

----c------ produces -------d------ } Normal level of Calcium in blood

9. Defects in hypothalamus may affect the activity of other endocrine glands.

Do you agree with this statement? Justify.

10. A decrease in the levels of a certain hormone in our body increases the amount of urine.

a) Identify this hormone.

b) Write a flowchart showing its action.

11. Ants follow definite paths while travelling because of certain chemicals they secrete to their surroundings.

a) Name this chemical.

b) Write any two other functions of these chemicals.

12. One of the hormones can be used for the treatment of allergy and asthma.

a) Which is the hormone mentioned above.

b) Which gland secretes this hormone?

c) Which are the other hormones secreted by this gland?

13. Write the probable reason for the actions of following farmers:

a) A farmer sprays Auxins on the crops.

b) A farmer sprays Gibberilins on the crops.

14. Blood test result of a person indicated Type-2 diabetes.

a) What is the cause of this disease?

b) Which gland is related to this disease?

c) Write the actions of hormone involved.

15. Complete the table with suitable titles.

Hints:- Low metabolic rate, Weight loss, Weight gain, Fatigue.

Heart beat increases, Heart beat slows down.

Title:	Title:
* * *	* * *

5. IMMUNITY AND HEALTHCARE

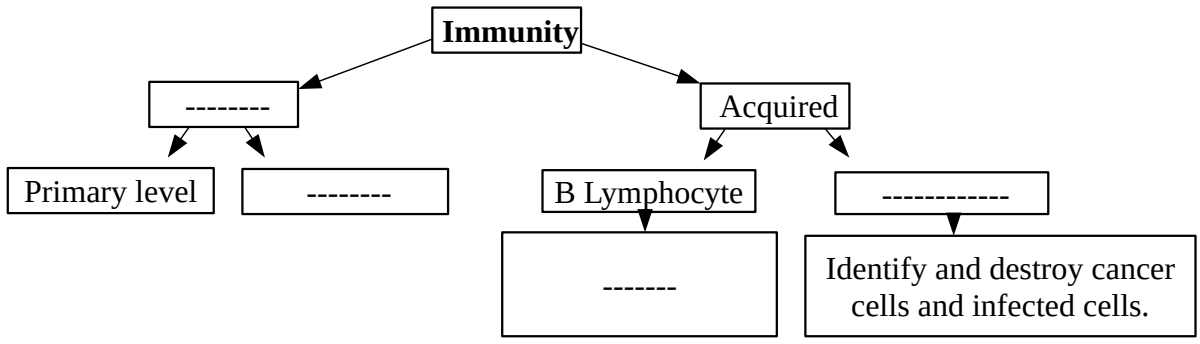
1. Choose and write the correct statement?
   a) Cilia in the respiratory tract prevent pathogens from entering the body
   b) HCl in the skin prevents pathogens
   c) Sebum in the stomach prevents pathogens
   d) Lysozyme found in the ear prevents pathogens

2. **Statement:** We live among pathogens, but we do not often get diseases.
   Reason: Our body has a strong immune system.

   a) Statement is true and reason false
   b) Statement and reason are true
   c) Statement and reason are false
   d) Statement is false because true

3. Some white blood cells can engulf and destroy pathogens.
   a) Name the process stated above.
   b) Which white blood cells are involved in this function?
   c) What are they called?

4. Complete the illustration:



5. Arrange the following stages involved in the process of blood clotting in right order?

   a) Red blood cells get trapped in the meshwork that makes fibrin fibers.
   b) Thromboplastin converts prothrombin into thrombin
   c) Platelets produce an enzyme called thromboplastin
   d) A blood clot is formed
   e) Thrombin converts fibrinogen into fibrin.

6. Immunization is an artificial way to prepare immune cells for the attack of a pathogen.
   a) Which vaccine is used to prevent cervical cancer?
   b) What are its components?

7. a. Should we be afraid of talking to or being close to an AIDS patient?
   Write your opinion
   b. List and explain the ways in which AIDS is transmitted?

8. * White spots in the mouth
   * Thick, discolored nails
   a) What is the disease indicated by these symptoms?
   b) Name the category of pathogen.

9. Select the one that matches in column A from column B

A	B
a. Produces chemicals against pathogens	Eosinophil
b. Destroys parasites	Basophil
c. Stimulates other immune cells	Neutrophil
d. Produces heparin.	Monocyte

10. Red blood cells become deformed.
   a) Which disease is this?
   b) What is the cause of this disease?
   c) How does it affect the body?

11.Sudden onset of fever, headache, muscle pain, red eyes, and fatigue
   a) Which disease shows these symptoms?
   b) Write the pathogen, the body part affected, and the method of transmission?

12. Antibiotics are the most effective medicine. However, they should be used as per the prescription of a doctor. Why?

13. Naveen's blood has A antigen, b antibody, and Rh
   Susheel's blood group is AB -ve
   a) What is Naveen's blood group?
   b) What are the antigens, antibodies, and Rh in Sushil's blood?

14. a) Brinjal – Blight - Bacteria
   b) Cassava - Mosaic- Fungus
   c) Coconut – Blight - Virus
   Which one is correct?

6. BIOLOGY AND TECHNOLOGY

Module 1. Man and Technology.

1. 'Along with the growth of humanity, science and technology has also developed'.
On the basis of above statement, arrange the following items into the respective boxes boxes:

- Synthetic biology
- Use of fire
- Beginning Agriculture
- Artificial Intelligence
- Cytology

Primitive man	**Modern man**
*	*
*	*
*	*
*	*

- Beginning social life
- DNA analysis
- Animal domestication

2. Choose some examples for genetic revolution from the given list:
[Hybrid plant varieties, Artificial meat, Disease resistant plant, Domestication of animals, DNA analysis to prove crime]

Module 2. Biotechnology and Genetic Engineering

3. Differentiate between biotechnology and genetic engineering.

4. The growth of biotechnology has accelerated by the discovery of ------- .
[Cell, Chromosome, DNA, Gene]

5. From the following items, select the two technologies used for genetic engineering:
- Recombinant DNA Technology
- DNA Fingerprinting.
- Artificial Intelligence
- CRISPR technology

6. "genetic scissor," "genetic glue," "gene vector," "host cell."
Using the above hints, briefly explain the process of Recombinant DNA Technology.

7. **CRISPER** technology is used for genetic engineering.
a). Who invented this technology?
b). What is **Cas9** in this technology?
c). What is the use of **gRNA** in CRISPR?

8. Fill in the table using the characteristics of the following Genetically modified microorganisms:

Psuedomonas putida	
E.coli bacterium	
Baker's yeast	
Rhizobium	

9. The technology used for the completion of Human Genome Project:
[Gene Mapping, Gene Therapy, Gene editing, Gene analysis]

10. What is Gene Therapy? How is this useful to mankind?

11. Mention any three application possibilities of DNA Finger printing.
Who proposed this?

12.



a). Identify and name the project indicated by this logo.
b). Where this project was launched?
c). 'This project has opened new avenues for diagnostic and treatment methods'.
Justify this statement.

13. Define:
    a). Genome b). Plasmid c). Junk DNA

Module 3. Biology and Artificial Intelligence (AI)

14. 'AI plays a major role in many areas of biological study'.
    Based on the hints given, substantiate this statement.
    Hints:- * Bioinformatics.
    * Diagnosing diseases.

Module 4. Technologies for conserving environment.

15. What is Cryopreservation? How is it useful to us?

16. Briefly explain the importance of Bioremediation.

Teachers participated in the creative workshop:

1. Najeeb Pothengodan, GHSS Kuzhimanna
2. Rasheed Odakkal, GVHSS Kondotty
3. Vaheeda K, GGVHSS Wandoor
4. Ramya C, CBHSS Vallikkunnu
5. Ragesh GHSS Anchachavidi


thank you